

1. Distinct Values from Employees Table

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
38 • INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_id)
39 values ("Raghav", "Male", 24, "2023-08-05", "Junior Associate", 25000.00, 1501),
40 ("Meera", "Female", 26, "2022-10-11", "Senior Associate", 30000.00, 1502),
41 ("Shiva", "Male", 30, "2021-02-10", "Manager", 40000.00, 1503),
42 ("Karthick", "Male", 32, "2020-06-12", "Senior Manager", 50000.00, 1504),
43 ("Ram", "Male", 34, "2019-03-09", "Zone General Manager", 70000.00, 1505) ;
44
45
46 • SELECT DISTINCT Employee_name, Designation, Salary
47 FROM Employees;
48
49 • SELECT Age As Employee_age FROM Employees;
```

The Result Grid shows the output of the first query:

Employee_name	Designation	Salary
Raghav	Junior Associate	25000.00
Meera	Senior Associate	30000.00
Shiva	Manager	40000.00
Karthick	Senior Manager	50000.00
Ram	Zone General Manager	70000.00

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
4	15:16:33	CREATE TABLE Departments (Department_id Int UNIQUE PRIMARY KEY, Department_name Varchar(40))	0 row(s) affected	0.047 sec
5	15:16:37	insert into Departments (Department_id, Department_name) values (1501, "Human Resource"), (1502, "R...)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
6	15:16:40	CREATE TABLE Location (Location_id Int UNIQUE PRIMARY KEY, Location_name Varchar(40))	0 row(s) affected	0.031 sec
7	15:16:44	CREATE TABLE Employees (Employee_id Int AUTO_INCREMENT UNIQUE PRIMARY KEY, Employee_...	0 row(s) affected	0.078 sec
8	15:16:49	INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
9	15:16:54	SELECT DISTINCT Employee_name, Designation, Salary FROM Employees LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec

2. ALIAS (AS)

Age changed to Employee_age

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
41 ("Shiva", "Male", 30, "2021-02-10", "Manager", 40000.00, 1503),
42 ("Karthick", "Male", 32, "2020-06-12", "Senior Manager", 50000.00, 1504),
43 ("Ram", "Male", 34, "2019-03-09", "Zone General Manager", 70000.00, 1505) ;
44
45
46 • SELECT DISTINCT Employee_name, Designation, Salary
47 FROM Employees;
48
49 • SELECT Age As Employee_age FROM Employees;
50 • SELECT Salary As Employee_salary FROM Employees;
51
52 • select * FROM employees
```

The Result Grid shows the output of the first query:

Employee_age
24
26
30
32
34

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
8	15:16:49	INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
9	15:16:54	SELECT DISTINCT Employee_name, Designation, Salary FROM Employees LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec
10	15:18:13	SELECT Age As Employee_age FROM Employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
11	15:18:13	SELECT Salary As Employee_salary FROM Employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	15:18:19	SELECT Age As Employee_age FROM Employees LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec
13	15:18:19	SELECT Salary As Employee_salary FROM Employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Salary changed to Employee_salary

File Edit View Query Database Server Tools Scripting Help

Navigator: Filter objects

SCHEMAS

- companydb
- dataset
- employee_details
- sale
- sales
- sys
- world

Administration Schemas

Information

No object selected

SQL File 4* SQL File 5* SQL File 6*

Limit to 1000 rows

```

41 ("Shiva", "Male", 30, "2021-02-10", "Manager", 40000.00, 1503),
42 ("Karthick", "Male", 32, "2020-06-12", "Senior Manager", 50000.00, 1504),
43 ("Ram", "Male", 34, "2019-03-09", "Zone General Manager", 70000.00, 1505);
44
45
46 SELECT DISTINCT Employee_name, Designation, Salary
47 FROM Employees;
48
49 SELECT Age As Employee_age FROM Employees;
50 SELECT Salary As Employee_salary FROM Employees;
51
52 select * FROM employees

```

Result Grid

Employee_salary
25000.00
30000.00
40000.00
50000.00
70000.00

Employees 4 Employees 5 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	15:16:49	INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_id, Location_id)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
9	15:16:54	SELECT DISTINCT Employee_name, Designation, Salary FROM Employees LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec
10	15:18:13	SELECT Age As Employee_age FROM Employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
11	15:18:13	SELECT Salary As Employee_salary FROM Employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	15:18:19	SELECT Age As Employee_age FROM Employees LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec
13	15:18:19	SELECT Salary As Employee_salary FROM Employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

3. WHERE CLAUSE & OPERATORS- Employees with Salary greater than 50000 and hired before 2023-08-10

File Edit View Query Database Server Tools Scripting Help

Navigator: Filter objects

SCHEMAS

- companydb
- dataset
- employee_details
- sale
- sales
- sys
- world

Administration Schemas

Information

No object selected

SQL File 4* SQL File 5* SQL File 6*

Limit to 1000 rows

```

47 FROM Employees;
48
49 SELECT Age As Employee_age FROM Employees;
50 SELECT Salary As Employee_salary FROM Employees;
51
52 select * FROM employees
53 where (Salary > 50000) AND (Hire_date < "2023-08-10");
54
55 select SUM(Salary) FROM Employees;
56
57 SELECT MIN(Age) FROM Employees;
58

```

Result Grid

Employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	Department_id	Location_id
5	Ram	Male	34	2019-03-09	Zone General Manager	70000.00	1505	1505

employees 6 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	15:16:54	SELECT DISTINCT Employee_name, Designation, Salary FROM Employees LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec
10	15:18:13	SELECT Age As Employee_age FROM Employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
11	15:18:13	SELECT Salary As Employee_salary FROM Employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	15:18:19	SELECT Age As Employee_age FROM Employees LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec
13	15:18:19	SELECT Salary As Employee_salary FROM Employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
14	15:20:09	select * FROM employees Where (Salary > 50000) AND (Hire_date < "2023-08-10") LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

3 B. Employee With missing Designation

SQL File 4*

```

38 INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_id)
39 values ("Raghav", "Male", 24, "2023-08-05", "Junior Associate", 25000.00, 1501),
40 ("Meera", "Female", 26, "2022-10-11", "Senior Associate", 30000.00, 1502),
41 ("Shiva", "Male", 30, "2021-02-10", "Manager", 40000.00, 1503),
42 ("Karthick", "Male", 32, "2020-06-12", NULL, 50000.00, 1504),
43 ("Ram", "Male", 34, "2019-03-09", "Zone General Manager", 70000.00, 1505);
44
45 select Employee_id, Employee_name, Gender, age, Designation, Salary From Employees
46 where Designation IS NULL;
47
48
49

```

Result Grid

Employee_id	Employee_name	Gender	age	Designation	Salary
4	Karthick	Male	32	NULL	50000.00

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:37:22	CREATE TABLE Employees (Employee_id INT AUTO_INCREMENT UNIQUE PRIMARY KEY, Employee_name VARCHAR(50), Gender VARCHAR(10), Age INT, Hire_date DATE, Designation VARCHAR(50), Salary DECIMAL(10,2), Department_id INT)	0 row(s) affected	0.078 sec
2	15:37:27	INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_id) values ("Raghav", "Male", 24, "2023-08-05", "Junior Associate", 25000.00, 1501), ("Meera", "Female", 26, "2022-10-11", "Senior Associate", 30000.00, 1502), ("Shiva", "Male", 30, "2021-02-10", "Manager", 40000.00, 1503), ("Karthick", "Male", 32, "2020-06-12", NULL, 50000.00, 1504), ("Ram", "Male", 34, "2019-03-09", "Zone General Manager", 70000.00, 1505);	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
3	15:37:36	Select Employee_id, Employee_name, Gender, age, Designation, Salary From Employees Where Designation IS NULL;	1 row(s) returned	0.000 sec / 0.000 sec
4	15:44:07	Select Employee_id, Employee_name, Gender, age, Designation, Salary From Employees Where Designation IS NULL;	1 row(s) returned	0.000 sec / 0.000 sec
5	15:44:07	Update Employees Set Designation = "Data Scientist" Where Employee_name = "Karthick"	Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE that ...	0.000 sec

Replacing NULL values in Designation with “Data Scientist” Where Emp ID is 4

SQL File 4*

```

42 ("Karthick", "Male", 32, "2020-06-12", NULL, 50000.00, 1504),
43 ("Ram", "Male", 34, "2019-03-09", "Zone General Manager", 70000.00, 1505);
44
45 select Employee_id, Employee_name, Gender, age, Designation, Salary From Employees
46 where Designation IS NULL;
47
48 Update Employees
49 Set Designation = "Data Scientist"
50 where Employee_id = 4;
51
52 select * from Employees;
53

```

Result Grid

Employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	Department_id	Location_id
1	Raghav	Male	24	2023-08-05	Junior Associate	25000.00	1501	NULL
2	Meera	Female	26	2022-10-11	Senior Associate	30000.00	1502	NULL
3	Shiva	Male	30	2021-02-10	Manager	40000.00	1503	NULL
4	Karthick	Male	32	2020-06-12	Data Scientist	50000.00	1504	NULL
5	Ram	Male	34	2019-03-09	Zone General Manager	70000.00	1505	NULL

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	15:37:36	Select Employee_id, Employee_name, Gender, age, Designation, Salary From Employees Where Designation IS NULL;	1 row(s) returned	0.000 sec / 0.000 sec
4	15:44:07	Select Employee_id, Employee_name, Gender, age, Designation, Salary From Employees Where Designation IS NULL;	1 row(s) returned	0.000 sec / 0.000 sec
5	15:44:07	Update Employees Set Designation = "Data Scientist" Where Employee_name = "Karthick"	Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE that ...	0.000 sec
6	15:46:20	Update Employees Set Designation = "Data Scientist" Where Employee_id = "Karthick"	Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE that ...	0.000 sec
7	15:47:13	Update Employees Set Designation = "Data Scientist" Where Employee_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
8	15:47:41	select * from Employees LIMIT 0.1000	5 row(s) returned	0.000 sec / 0.000 sec

SORTING AND GROUPING

1. ORDER BY- Employees sorted based on Department_id in ascending order

SQL File 4* SQL Assignment-Distinct Alias... SQL File 5* SQL File 6*

```

14
15
16 • INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_id)
17 values ('Raghav', 'Male', 24, '2023-08-05', 'Junior Associate', 25000.00, 1501),
18 ('Meera', 'Female', 26, '2022-10-11', 'Senior Associate', 30000.00, 1502),
19 ('Shiva', 'Male', 30, '2021-02-10', 'Manager', 40000.00, 1503),
20 ('Karthick', 'Male', 32, '2020-06-12', 'Data Scientist', 50000.00, 1504),
21 ('Ram', 'Male', 34, '2019-03-09', 'Zone General Manager', 70000.00, 1505);
22
23 • Select * From Employees
24 ORDER BY Department_id;
25
--

```

Employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	Department_id
1	Raghav	Male	24	2023-08-05	Junior Associate	25000.00	1501
2	Meera	Female	26	2022-10-11	Senior Associate	30000.00	1502
3	Shiva	Male	30	2021-02-10	Manager	40000.00	1503
4	Karthick	Male	32	2020-06-12	Data Scientist	50000.00	1504
5	Ram	Male	34	2019-03-09	Zone General Manager	70000.00	1505

Employees 15 x

Output

#	Time	Action	Message	Duration / Fetch
8	16:07:50	Select * From Employees ORDER BY Department_id, Salary Desc LIMIT 0. 1000	5 row(s) returned	0.000 sec / 0.000 sec
9	16:11:25	Select * From Employees ORDER BY Salary Desc, Department_id LIMIT 0. 1000	5 row(s) returned	0.000 sec / 0.000 sec
10	16:12:30	Select * From Employees ORDER BY Salary Desc LIMIT 0. 1000	5 row(s) returned	0.000 sec / 0.000 sec
11	16:12:49	Select * From Employees ORDER BY Department_id LIMIT 0. 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	16:13:10	Select * From Employees ORDER BY Department_id, Salary Desc LIMIT 0. 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	16:14:15	Select * From Employees ORDER BY Department_id LIMIT 0. 1000	5 row(s) returned	0.000 sec / 0.000 sec

Employees sorted based on Salary in Descending order

SQL File 4* SQL Assignment-Distinct Alias... SQL File 5* SQL File 6*

```

15
16 • INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_id)
17 values ('Raghav', 'Male', 24, '2023-08-05', 'Junior Associate', 25000.00, 1501),
18 ('Meera', 'Female', 26, '2022-10-11', 'Senior Associate', 30000.00, 1502),
19 ('Shiva', 'Male', 30, '2021-02-10', 'Manager', 40000.00, 1503),
20 ('Karthick', 'Male', 32, '2020-06-12', 'Data Scientist', 50000.00, 1504),
21 ('Ram', 'Male', 34, '2019-03-09', 'Zone General Manager', 70000.00, 1505);
22
23 • Select * From Employees
24 ORDER BY Salary Desc;
25
--

```

Employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	Department_id
5	Ram	Male	34	2019-03-09	Zone General Manager	70000.00	1505
4	Karthick	Male	32	2020-06-12	Data Scientist	50000.00	1504
3	Shiva	Male	30	2021-02-10	Manager	40000.00	1503
2	Meera	Female	26	2022-10-11	Senior Associate	30000.00	1502
1	Raghav	Male	24	2023-08-05	Junior Associate	25000.00	1501

Employees 16 x

Output

#	Time	Action	Message	Duration / Fetch
9	16:11:25	Select * From Employees ORDER BY Salary Desc, Department_id LIMIT 0. 1000	5 row(s) returned	0.000 sec / 0.000 sec
10	16:12:30	Select * From Employees ORDER BY Salary Desc LIMIT 0. 1000	5 row(s) returned	0.000 sec / 0.000 sec
11	16:12:49	Select * From Employees ORDER BY Department_id LIMIT 0. 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	16:13:10	Select * From Employees ORDER BY Department_id, Salary Desc LIMIT 0. 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	16:14:15	Select * From Employees ORDER BY Department_id LIMIT 0. 1000	5 row(s) returned	0.000 sec / 0.000 sec
14	16:16:01	Select * From Employees ORDER BY Salary Desc LIMIT 0. 1000	5 row(s) returned	0.016 sec / 0.000 sec

2. Displayed Employees hired in year 2018 using LIMIT

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- companydb
- dataset
- employee_details
- sale
- sales
- sys
- world

Administration Schemas

Information

No object selected

SQL File 4* SQL Assignment-Distinct, Alias... SQL File 5* SQL File 6*

Limit to 1000 rows

```

20 ("Karthick", "Male", 32, "2018-06-12", "Data Scientist", 50000.00, 1504),
21 ("Ram", "Male", 34, "2018-03-09", "Zone General Manager", 70000.00, 1505),
22 ("Ezhil", "Male", 23, "2020-08-07", "Junior Associate", 25000.00, 1506),
23 ("Suresh", "Male", 26, "2021-10-11", "Senior Associate", 29000.00, 1507);
24
25 ##LIMIT
26
27 • select * from Employees
28 where Hire_date > 2018
29 LIMIT 5;
30
31

```

Result Grid

Employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	Department_id
1	Raghav	Male	24	2018-09-05	Junior Associate	25000.00	1501
2	Meera	Female	26	2018-10-11	Senior Associate	30000.00	1502
3	Shiva	Male	30	2018-02-10	Manager	40000.00	1503
4	Karthick	Male	32	2018-06-12	Data Scientist	50000.00	1504
5	Ram	Male	34	2018-03-09	Zone General Manager	70000.00	1505

Employees 35 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	16:29:49	Select * from Employees Where Hire_date > 2018 LIMIT 5	5 row(s) returned	0.015 sec / 0.000 sec
15	16:29:59	Select * from Employees Where Hire_date > 2018 LIMIT 5	5 row(s) returned	0.000 sec / 0.000 sec
16	16:30:13	Select * from Employees Where Hire_date < 2018 LIMIT 5	0 row(s) returned	0.000 sec / 0.000 sec
17	16:30:20	Select * from Employees Where Hire_date > 2018 LIMIT 5	5 row(s) returned	0.000 sec / 0.000 sec
18	16:30:34	Select * from Employees Where Hire_date > 2018 LIMIT 3	3 row(s) returned	0.000 sec / 0.000 sec
19	16:30:42	Select * from Employees Where Hire_date > 2018 LIMIT 5	5 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

3. Aggregate Functions- Sum of All Salaries

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- companydb
- dataset
- employee_details
- sale
- sales
- sys
- world

Administration Schemas

Information

No object selected

SQL File 4* SQL Assignment-Distinct, Alias... SQL File 5* SQL File 6*

Limit to 1000 rows

```

33 ##LIMIT
34
35 • Select * from Employees
36 where Hire_date > 2018
37 LIMIT 5;
38
39 ## Aggregate Functions
40
41 • Select SUM(Salary) From Employees;
42
43
44

```

Result Grid

SUM(Salary)
269000.00

Result 36 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
15	16:29:59	Select * from Employees Where Hire_date > 2018 LIMIT 5	5 row(s) returned	0.000 sec / 0.000 sec
16	16:30:13	Select * from Employees Where Hire_date < 2018 LIMIT 5	0 row(s) returned	0.000 sec / 0.000 sec
17	16:30:20	Select * from Employees Where Hire_date > 2018 LIMIT 5	5 row(s) returned	0.000 sec / 0.000 sec
18	16:30:34	Select * from Employees Where Hire_date > 2018 LIMIT 3	3 row(s) returned	0.000 sec / 0.000 sec
19	16:30:42	Select * from Employees Where Hire_date > 2018 LIMIT 5	5 row(s) returned	0.000 sec / 0.000 sec
20	16:34:46	Select SUM(Salary) From Employees LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

- Minimum age of Employees

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

35 Select * from Employees
36 Where Hire_date > 2018
37 LIMIT 5;
38
39 ## Aggregate Funtions
40
41 Select SUM(Salary) From Employees;
42
43 Select MIN(Age) From Employees;
44
45
46

```

The Results tab shows the output of the last query (line 43):

MIN(Age)
23

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
16	16:30:13	Select * from Employees Where Hire_date < 2018 LIMIT 5	0 row(s) returned	0.000 sec / 0.000 sec
17	16:30:20	Select * from Employees Where Hire_date > 2018 LIMIT 5	5 row(s) returned	0.000 sec / 0.000 sec
18	16:30:34	Select * from Employees Where Hire_date > 2018 LIMIT 3	3 row(s) returned	0.000 sec / 0.000 sec
19	16:30:42	Select * from Employees Where Hire_date > 2018 LIMIT 5	5 row(s) returned	0.000 sec / 0.000 sec
20	16:34:46	Select SUM(Salary) From Employees LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
21	16:36:09	Select MIN(Age) From Employees LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

GROUP BY

1. Maximum salary for each locations

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

40 ## Aggregate Funtions
41
42 Select SUM(Salary) From Employees;
43
44 Select MIN(Age) From Employees;
45
46 ## GROUP BY
47
48 Select Location, Max(Salary) From Employees
49 Group By Location;
50
51

```

The Results tab shows the output of the last query (line 48):

Location	Max(Salary)
Chennai	40000.00
Bangalore	70000.00
Mumbai	29000.00

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	16:44:12	SELECT Location, Max(Salary) From Employees GROUP BY Location LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec

2. Average Salary For Each designation Where word is analyst

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

44
45 • Select MIN(Age) From Employees;
46
47 ## GROUP BY
48
49 • SELECT Location, Max(Salary) From Employees
50 GROUP BY Location;
51
52 • Select Designation, AVG(Salary) As Analyst_Salary From Employees
53 Where Designation = "Analyst"
54 GROUP BY Designation;
55

```

The Results tab shows the output of the query:

Designation	Analyst_Salary
Analyst	26333.333333

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	16:59:58	CREATE TABLE Employees (Employee_id Int AUTO_INCREMENT UNIQUE PRIMARY KEY, Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_id, ...)	0 row(s) affected	0.078 sec
2	17:00:27	INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_id, ...)	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
3	17:01:01	Select Designation, AVG(Salary) As Analyst_Salary From Employees Where Designation = "Analyst" GROUP BY Designation	1 row(s) returned	0.000 sec / 0.000 sec

HAVING:

1. Department with less than 3 employees

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

50 GROUP BY Location;
51
52 • Select Designation, AVG(Salary) As Analyst_Salary From Employees
53 Where Designation = "Analyst"
54 GROUP BY Designation;
55
56 ## HAVING
57
58 • Select Department, Count(Employee_name) As Employee_count From Employees
59 GROUP BY Department Having Employee_Count < 3;
60
61

```

The Results tab shows the output of the query:

Department	Employee_count
Human Resource	1
IT	1
Insurance	1

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
2	17:00:27	INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_id, ...)	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
3	17:01:01	Select Designation, AVG(Salary) As Analyst_Salary From Employees Where Designation = "Analyst" GROUP BY Designation	1 row(s) returned	0.000 sec / 0.000 sec
4	17:02:01	Drop Table Employees	0 row(s) affected	0.047 sec
5	17:02:21	CREATE TABLE Employees (Employee_id Int AUTO_INCREMENT UNIQUE PRIMARY KEY, Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_id, ...)	0 row(s) affected	0.063 sec
6	17:05:22	INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_id, ...)	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.016 sec
7	17:11:22	Select Department, Count(Employee_name) As Employee_count From Employees GROUP BY Department Having Employee_Count < 3	3 row(s) returned	0.000 sec / 0.000 sec

2. Locations with Female employees whose avg age is 30

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 4* SQL Assignment- Distinct Alias SQL File 5* SQL File 6*

SCHMAS: Filter objects companydb dataset employee_details sale sales sys world

Administration Schemas

Information: No object selected

Result Grid: Location Avg_Female_Age
Chennai 25
Mumbai 25

SQL File 4* SQL Assignment- Distinct Alias SQL File 5* SQL File 6*

```

53 Where Designation = "Analyst"
54 GROUP BY Designation;
55
56 ## HAVING
57
58 Select Department, Count(Employee_name) As Employee_count From Employees
59 GROUP BY Department Having Employee_Count < 3;
60
61 Select Location, ROUND(Avg(Age), 0) AS Avg_Female_Age From Employees
62 Where Gender = "Female"
63 GROUP BY Location;
64

```

Output: Action Output

#	Time	Action	Message	Duration / Fetch
3	17:01:01	Select Designation, AVG(Salary) As Analyst_Salary From Employees Where Designation = "Analyst" GRO...	1 row(s) returned	0.000 sec / 0.000 sec
4	17:02:01	Drop Table Employees	0 row(s) affected	0.047 sec
5	17:02:21	CREATE TABLE Employees (Employee_id Int AUTO_INCREMENT UNIQUE PRIMARY KEY, Employee_...	0 row(s) affected	0.063 sec
6	17:05:22	INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department_J...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.016 sec
7	17:11:22	Select Department, Count(Employee_name) From Employees GROUP BY Departme...	3 row(s) returned	0.000 sec / 0.000 sec
8	17:15:35	Select Location, ROUND(Avg(Age), 0) AS Avg_Female_Age From Employees Where Gender = "Female"...	2 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

JOINS

INNER JOIN

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 4* SQL Assignment- Window funci... SQL File 4* SQL File 5* SQL File 6*

SCHMAS: Filter objects companydb dataset employee_details sale sales sys world

Administration Schemas

Information: No object selected

Result Grid: Employee_name Designation Department
Raghav Junior Analyst Insurance
Meena Junior Analyst Insurance
Krishna Senior Analyst Human Resource
Ram Assistant Manager Finance
Karthick Manager Corporate
Reena Senior Analyst IT

SQL File 4* SQL Assignment- Window funci... SQL File 4* SQL File 5* SQL File 6*

```

48 ('Krishna', 'Male', 'Senior Analyst', 30000.00, 102, 'Human Resource'),
49 ('Ram', 'Male', 'Assistant Manager', 45000.00, 103, 'Finance'),
50 ('Karthick', 'Male', 'Manager', 50000.00, 104, 'Corporate'),
51 ('Reena', 'Female', 'Senior Analyst', 35000.00, 105, 'IT') ;
52
53 ## Inner JOIN
54
55 Select Employee_name, Designation, Department
56 FROM Employees
57 INNER JOIN Departments
58 ON Employees.Department_id = Departments.Department_id;
59

```

Output: Action Output

#	Time	Action	Message	Duration / Fetch
1	16:29:48	Select Employee_name, Designation, Department FROM Employees INNER JOIN Departments ON Emplo...	6 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

LEFT JOIN

Local instance MySQL80 x unconnected x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

- companydb
- dataset
- employee_details
- sale
- sales
- sys
- world

Administration Schemas

Information

No object selected

SQL File 4*

```

56 FROM Employees
57 INNER JOIN Departments
58 ON Employees.Department_id = Departments.Department_id;
59
60 ## LEFT JOIN
61
62 • Select Department, Employee_name
63 FROM Employees
64 LEFT JOIN Departments
65 ON Employees.Department_id = Departments.Department_id;
66
67

```

Result Grid

Department	Employee_name
Insurance	Raghav
Insurance	Meena
Human Resource	Krishna
Finance	Ram
Corporate	Karthick
IT	Reena

Result 4 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	16.35.46	Select Department, SUM(Employee_name) AS Employee_count FROM Employees LEFT JOIN Departme...	Error Code: 1140. In aggregated query without GROUP BY, expression #1 of SELECT list contains nonaggr...	0.016 sec
2	16.37.33	Select Department, Employee_name FROM Employees LEFT JOIN Departments ON Employees Departme...	6 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

RIGHT JOIN

Local instance MySQL80 x unconnected x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

- companydb
- dataset
- employee_details
- sale
- sales
- sys
- world

Administration Schemas

Information

No object selected

SQL File 4*

```

66 • Select Department, Employee_name
67 FROM Employees
68 LEFT JOIN Departments
69 ON Employees.Department_id = Departments.Department_id;
70
71 ##RIGHT JOIN
72
73 • Select Employee_name, Location_name
74 From Location
75 RIGHT JOIN Employees
76 ON Employees.Location_id = Location.Location_id;
77

```

Result Grid

Employee_name	Location_name
Meena	Delhi
Krishna	Mumbai
Ram	Pune
Karthick	Bangalore
Reena	Bangalore

Result 10 x

Output

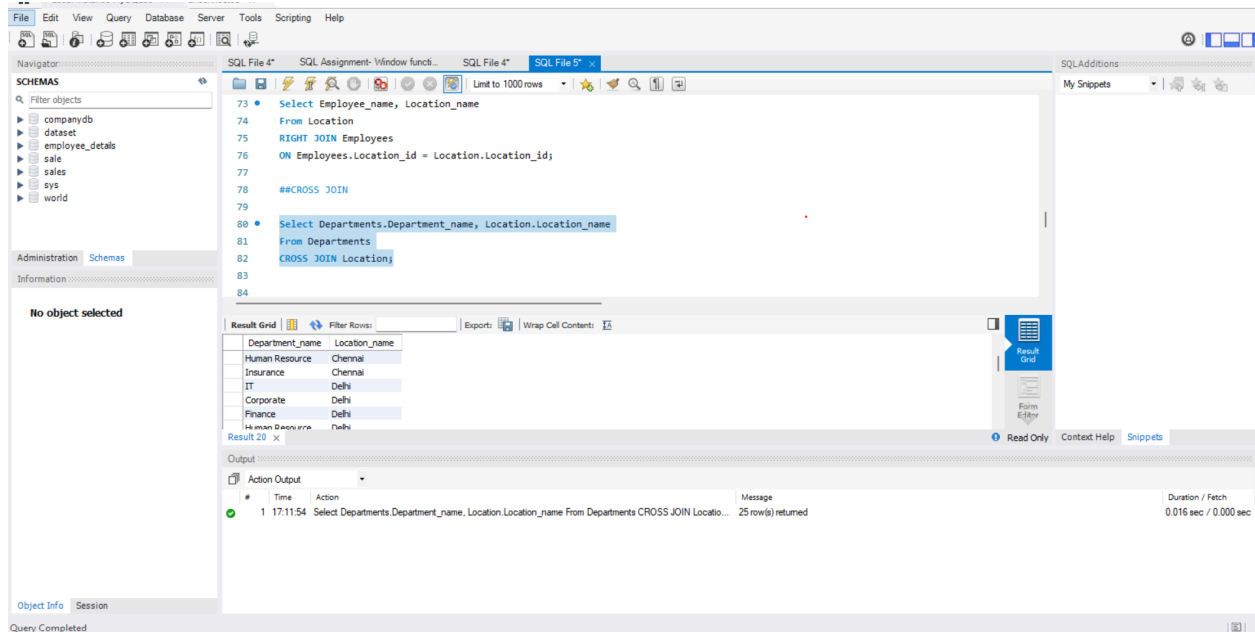
Action Output

#	Time	Action	Message	Duration / Fetch
13	16.43.10	Select Employee_name, Location_name From Employees RIGHT JOIN Location ON Employees Location...	6 row(s) returned	0.000 sec / 0.000 sec
14	16.43.27	Drop Table Employees	0 row(s) affected	0.046 sec
15	16.43.32	CREATE TABLE Employees (Employee_id INT AUTO_INCREMENT UNIQUE PRIMARY KEY, Employee...	0 row(s) affected	0.110 sec
16	16.43.36	INSERT INTO Employees (Employee_name, Gender, Designation, Salary, Department_id, Department...	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0	0.016 sec
17	16.43.40	Select Employee_name, Location_name From Employees RIGHT JOIN Location ON Employees Location...	5 row(s) returned	0.000 sec / 0.000 sec
18	16.50.34	Select Employee_name, Location_name From Location RIGHT JOIN Employees ON Employees Location...	6 row(s) returned	0.000 sec / 0.000 sec

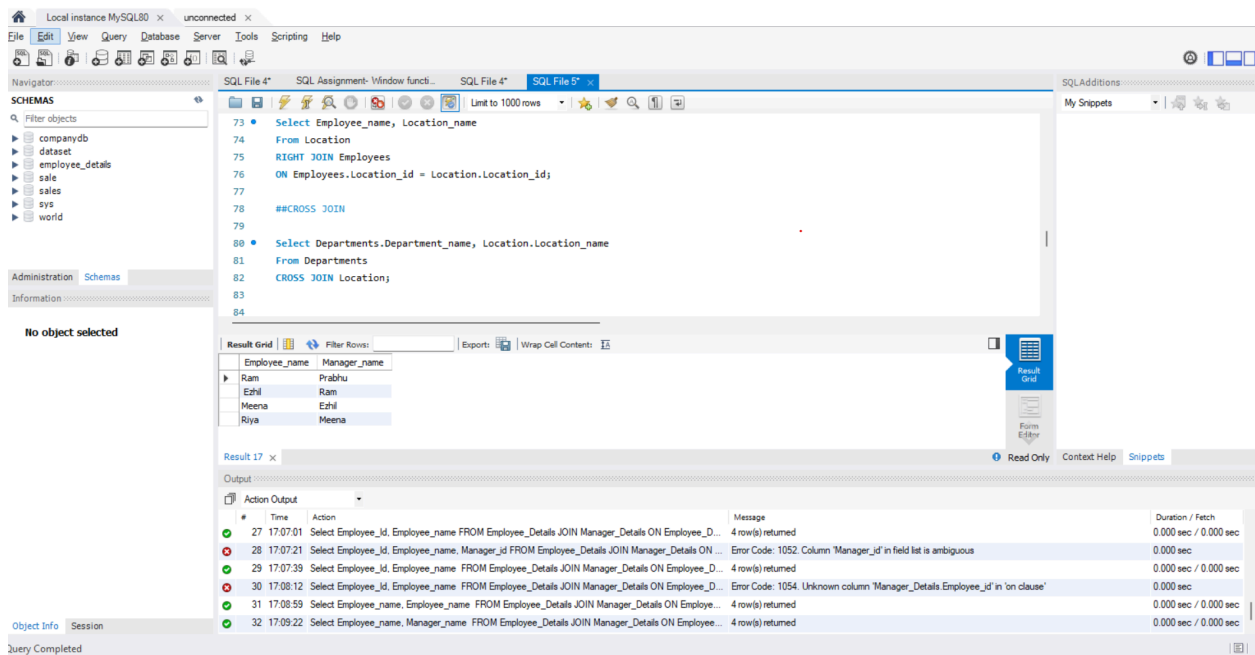
Object Info Session

Query Completed

CROSS JOIN



SELF JOIN



WINDOWS FUNCTIONS

1. Window function query to rank employees by salary using rank().

The screenshot shows the SQL Developer interface with a query window containing the following SQL code:

```

46 ("Alice", "Female", 26, "2021-10-11", "Analyst", 29000.00, "Insurance");
47
48 ## Rank Employee Based on Salary
49
50 • Select
51 Employee_name,
52 Salary,
53 RANK() OVER(ORDER BY Salary) Employee_Rank
54 FROM Employees;
55
56
57

```

The result grid displays the following data:

Employee_name	Salary	Employee_Rank
Raghav	25000.00	1
Roshni	26000.00	2
Alice	29000.00	3
Meera	30000.00	4
Shiva	40000.00	5

The output pane shows the execution log with the following messages:

```

# Time Action Message Duration / Fetch
1 13:24:13 CREATE TABLE Employees ( Employee_id INT AUTO_INCREMENT UNIQUE PRIMARY KEY, Employee_... 0 row(s) affected 0.062 sec
2 13:24:21 INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department) val... 7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0 0.000 sec
3 13:24:25 Select Employee_name, Salary, RANK() OVER(ORDER BY Salary) Employee_Rank FROM Employees 7 row(s) returned 0.000 sec / 0.000 sec

```

2. Write a window function query to rank employees by salary within each department Using DENSE RANK()

The screenshot shows the SQL Developer interface with a query window containing the following SQL code:

```

52 Salary,
53 RANK() OVER(ORDER BY Salary) Employee_Rank
54 FROM Employees;
55
56 ## Rank of employee within Each department using Dense Rank
57 • Select
58 Employee_name,
59 Salary,
60 Department,
61 DENSE_RANK() OVER(PARTITION BY Department ORDER BY Salary) Rank_status
62 FROM Employees;
63

```

The result grid displays the following data:

Employee_name	Salary	Department	Rank_status
Ram	70000.00	Corporate	1
Raghav	25000.00	Finance	1
Meera	30000.00	Finance	2
Shiva	40000.00	Finance	3
Karthick	50000.00	Human Resource	1

The output pane shows the execution log with the following messages:

```

# Time Action Message Duration / Fetch
1 13:24:13 CREATE TABLE Employees ( Employee_id INT AUTO_INCREMENT UNIQUE PRIMARY KEY, Employee_... 0 row(s) affected 0.062 sec
2 13:24:21 INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department) val... 7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0 0.000 sec
3 13:24:25 Select Employee_name, Salary, RANK() OVER(ORDER BY Salary) Employee_Rank FROM Employees 7 row(s) returned 0.000 sec / 0.000 sec
4 13:27:58 Select Employee_name, Salary, Department, DENSE_RANK() OVER(PARTITION BY Department ORDER... 7 row(s) returned 0.016 sec / 0.000 sec

```

3. Write a window function query, Running total salary by department

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- companydb
- dataset
- employee_details
- sale
- sales
- sys
- world

Administration Schemas

Information

No object selected

SQL File 4* SQL File 2* SQL File 4*

Limit to 1000 rows

SQL Additions

My Snippets

```
58 Employee_name,
59 Salary,
60 Department,
61 DENSE_RANK() OVER(PARTITION BY Department ORDER BY Salary) Rank_status
62 FROM Employees;
63
64
65 • Select Employee_name, Department,
66 SUM(Salary) OVER (PARTITION BY Department ORDER BY salary) As Running_total
67 FROM Employees;
68
69
---
```

Result Grid

Employee_name	Department	Running_total
Ram	Corporate	70000.00
Raghav	Finance	25000.00
Meera	Finance	50000.00
Shiva	Finance	50000.00
Karthick	Human Resoure	50000.00

Result 12 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	13:24:13	CREATE TABLE Employees (Employee_id Int AUTO_INCREMENT UNIQUE PRIMARY KEY, Employee_name, Gender, Age, Hire_date, Designation, Salary, Department) val...	0 row(s) affected	0.062 sec
2	13:24:21	INSERT INTO Employees (Employee_name, Gender, Age, Hire_date, Designation, Salary, Department) val...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
3	13:24:25	Select Employee_name, Salary, RANK() OVER(ORDER BY Salary) Employee_Rank FROM Employees	7 row(s) returned	0.000 sec / 0.000 sec
4	13:27:58	Select Employee_name, Salary, Department, DENSE_RANK() OVER(PARTITION BY Department ORDER...	7 row(s) returned	0.016 sec / 0.000 sec
5	13:37:51	Select SUM(Salary) OVER (PARTITION BY Department ORDER BY salary) As Running_total FROM Empl...	7 row(s) returned	0.000 sec / 0.000 sec
6	13:38:31	Select Employee_name, Department, SUM(Salary) OVER (PARTITION BY Department ORDER BY salary) ...	7 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed